

# Krotov: A Python implementation of Krotov's method for quantum optimal control

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## ABSTRACT

We present a new open-source Python package, *Krotov*, implementing the quantum optimal control method of that name. It allows us to determine time-dependent external fields for a wide range of quantum control problems, including state-to-state transfer, quantum gate implementation and optimization towards an arbitrary performance measure. Krotov's method compares to other gradient-based optimization methods such as gradient ascent and guarantees monotonic convergence to approximately time-optimum control fields. The user-friendly interface allows easy combination with other Python packages, and thus high-level customizations.

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